Caffeine Consumption: Survey Analysis

Portfolio case study | Jane Okon-Jeffrey | University of Houston–Downtown | STAT 6312 Biostatistics (2024)

Project overview

This coursework project explored self-reported beverage preferences, reasons for consumption, daily intake, and perceived sleep effects. An online Google Forms survey was shared with family and friends. The final dataset contains **25 responses**. The analysis uses descriptive statistics and conditional proportions to examine patterns in this small convenience sample.

Data and approach

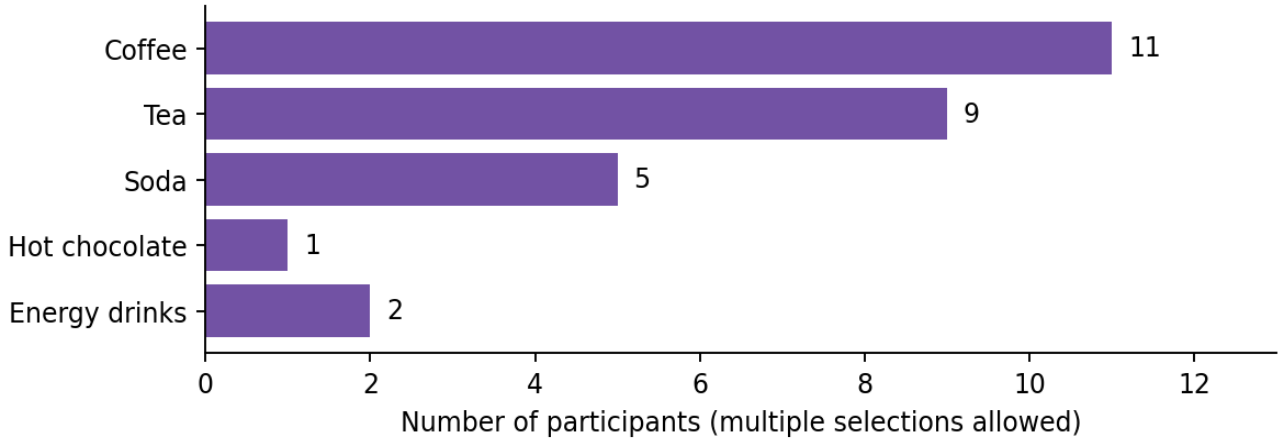
The original Excel workbook, *Caffeine Consumption (Responses)*, was used to verify the published figures. Responses were summarized using frequencies, percentages, multi-select counts, and cross-tabulations. No statistical significance or causal effects are claimed. “Regular beverage selection” and “more than zero drinks reported per day” are treated as separate measures.

Sample at a glance

Measure	Verified result
Participants	25
Sex	15 female (60%); 10 male (40%)
Selected at least one listed beverage	18 of 25 (72%)
Reported more than zero drinks per day	17 of 25 (68%)
Reported no daily drinks	8 of 25 (32%)

Beverage choices

Coffee was the most commonly selected beverage (11 of 25; 44%), followed by tea (9 of 25; 36%). Participants could select more than one beverage, so these percentages do not sum to 100%.



Motivations and perceived sleep effects

Taste was selected by 16 participants (64%), increased energy/alertness by 10 (40%), enhanced mood by 8 (32%), and social reasons by 5 (20%). These are multiple-response items. For sleep, 10 respondents (40%) reported no effect, 7 (28%) slight effects, 2 (8%) significant effects, and 6 (24%) were unsure.

Daily intake and cross-checks

Reported drinks per day	Participants	Share
0	8	32%
1	15	60%
2	2	8%

Among respondents selecting at least one listed beverage, 13 were female and 5 were male. This corrects an inconsistent gender count in the original coursework report. Of the 17 respondents reporting more than zero drinks per day, 2 reported significant sleep effects: **2/17 = 11.8%**. This is a descriptive conditional proportion, not evidence that caffeine caused those effects.

Data quality notes

Two respondents selected “None of the above” as their regular beverage choice but also reported one drink per day. The original responses have been preserved; these inconsistent answers explain why the two measures differ. Selecting a listed beverage is not a clinical or population-level definition of caffeine use.

Interpretation and limitations

The results illustrate common beverage choices and motivations within this group, rather than patterns for the general public. The survey used a small, non-random convenience sample, self-reported answers, and a single time point. It cannot establish cause and effect, incidence over time, or statistically significant differences between age or sex groups without further analysis.

What this project demonstrates

Survey data preparation and validation; categorical summaries and percentages; handling multiple-response questions; descriptive conditional probability; charting; and transparent reporting of inconsistent data and study limitations.

Source and methodology note

Source: Jane Okon-Jeffrey, *Caffeine Consumption (Responses).xlsx*, 25 original survey responses; companion coursework report, *Caffeine Consumption: The Patterns, Motivations, and Effects Across Different Demographics* (June 25, 2024). All figures above were checked against the original response worksheet. This portfolio summary corrects selected tabulation and interpretation issues; the original coursework submission is unchanged.